

NORMIA

AEC Regulatory Intelligence & Compliance Consultant

Concept Design Compliance Review

Villa Samaa Al Bahr

Project	Villa Samaa Al Bahr
Location	Qetaifan Island North - East Villa Precinct, Al Daayen, State of Qatar
Plot	QN-V2-B-15 (land area 786.15 m2 as stated on the drawings)
Client	Mohammad Jaber
Architect	NAZA Architecture Studio / grown studio, Mexico City
Consultant	Nemanja Biondic
Design stage	S.D - Schematic Design
Drawings reviewed	12 sheets (A-100 to A-104, A-200 to A-202, A-300 to A-303), dated 18 May / 13 June 2026
Review date	10 August 2026
Output mode	CONCEPT DESIGN REVIEW

This is a demonstration of the NORMIA review protocol. Findings are drawn from verified regulation records: the Plot Building Regulation Sheet for QN-V2-B-15, the Qetaifan Island North Planning & Design Guidelines Rev 2 (May 2023), the Front Boundary Wall document Rev 00 (November 2025), and the Qatar national regulation library. Where a control could not be verified, the finding says so rather than assuming.

1. Summary

Severity	Count	Meaning
CRITICAL	4	Affects legal approval. Massing-level change required.
HIGH	4	Likely authority rejection or major redesign.
MEDIUM	5	Design modification required; concept survives.
INFORMATION REQUIRED	5	Cannot be determined from the submitted set.
VERIFIED COMPLIANT	6	Tested and passed against a verified control.

Overall compliance: not meaningful

A percentage would misrepresent the position. Two critical findings go to the permitted quantum and storey count of the building, and the set contains no site plan - so setback compliance cannot be tested in either direction. NORMIA does not issue a compliant result where the governing evidence is absent.

The finding that governs the scheme

The drawings state a **BUILT AREA of 819.65 m2 against a maximum permitted GFA of 472 m2 for this plot** - an apparent excess of 347.65 m2, roughly 174% of the allowance. The basement cannot be used to absorb it: as drawn it contains bedrooms, bathrooms, a kitchen and living space, and the Design Guidelines exclude basement area from GFA only where it is BOTH naturally ventilated AND non-habitable.

Separately, the building does not fit inside its own setback envelope. Computed from the approved plot corner coordinates, the buildable envelope is 27.14 m long. The first floor is dimensioned at 30.00 m overall. It is over by 2.86 m - and that is independent of where the building is positioned on the plot.

2. Compliance Register

Each row cites the regulation record it was tested against. Statuses: CRITICAL / HIGH / MEDIUM / INFO REQUIRED / COMPLIANT.

ID	Requirement	Design condition	Status	Source
C-01	Max GFA 472 m2 for this plot. Ancillary buildings count within it.	BUILT AREA 819.65 m2 stated on every sheet. No area schedule; composition not stated.	CRITICAL	Plot Building Regulation Sheet; QIN DG Rev 2 p.92, p.98
C-02	G+1+P only. Penthouse limited to 40% of first-floor footprint, min 3 m back from first-floor front and rear facades.	A-103 is titled 'Second floor' - a full storey of bedrooms, living, kitchen and terraces. Sections label the same level 'penthouse'.	CRITICAL	Plot sheet; QIN DG Rev 2 p.92, p.93 (Key Param G)
H-01	DG level bands: ground floor max +5.5 m; first floor max +9.0 m.	First floor +6.825 (over by 1.325 m). Penthouse +10.675 (over by 1.675 m).	HIGH	QIN DG Rev 2 p.95 (Sheet 4/7)
H-02	Max 15.00 m measured FROM THE LOWEST SIDE OF THE PLOT, including all parapets and roof structures.	Building tops at +15.00 measured from 'access sidewalk' +/-0.00. Wrong datum, and zero headroom remains.	HIGH	QIN DG Rev 2 p.93 (Key Param B)
C-03	Building must sit within the setback envelope: 5 m front / 3 m sides / 6 m rear.	Buildable envelope computed from the approved plot coordinates is 27.14 x 15.85 m. First floor is dimensioned 30.00 m overall (26.00 m grid plus 2.00 m projections at each end). OVER BY 2.86 m in length, at any position.	CRITICAL	Plot sheet coordinates; QIN DG Rev 2 p.92-93
C-04	Basement excluded from GFA only if BOTH naturally ventilated AND non-habitable. Habitable area max 50% of basement footprint.	A-100 shows bedrooms, bathrooms, a kitchen and living space at basement level. A substantial part of the basement is habitable and therefore counts toward the 472 m2 GFA.	CRITICAL	QIN DG Rev 2 p.94 (Key Param F), p.95
H-04	Max 15.00 m includes all parapets and roof structures; screens concealing roof structures sit BELOW the 15 m line.	A-104 annotates roof terrace levels of FFL +15.00 and +14.75 with a pergola at RL +14.525/+15.00. An occupied floor AT the limit cannot carry a parapet or balustrade without exceeding it.	HIGH	QIN DG Rev 2 p.93 (Key Param B), p.95
H-03	Setbacks 5 m front (max 10 m) / 3 m sides / 6 m rear must be demonstrable.	No site plan in the set. No sheet dimensions the building to the plot boundary.	HIGH	Plot sheet; QIN DG Rev 2 p.92, p.93 (Key Param F)
M-01	Only architectural elements may project 1.5 m into the neighbour side setback, and they may carry no openings.	First floor shows 2.00 m projections beyond grids 1 and 5 (overall 30.00 m against a 26.00 m grid).	MEDIUM	QIN DG Rev 2 p.92, p.93 (Key Param H)
M-02	Basement must follow the BUILDING SETBACK, not the plot boundary.	Basement outline on A-100 appears to extend beyond the above-ground footprint toward the east boundary. Not dimensioned.	MEDIUM	QIN DG Rev 2 p.92, p.94
M-03	Boundary wall height - 2.50 m front (Nov 2025) versus 3.00 m (plot sheet / DG). Conflict unresolved.	No boundary wall height annotated on any elevation.	MEDIUM	Front Boundary Wall Rev 00; NORMIA conflict record

ID	Requirement	Design condition	Status	Source
M-04	Minimum 2 parking spaces. Lusail typical bay 2.65 x 5.80 m, aisle 6.70 m.	CORRECTED ON REVIEW: A-100 shows 4 numbered bays plus a 10% ramp - the count exceeds the 2 minimum. Bay dimensions and aisle width remain unannotated.	MEDIUM	Plot sheet; QIN DG Rev 2 p.28
M-05	Roof terrace permitted only where at least 6 m from side AND rear plot boundaries, or adequately screened.	A-104 shows an extensive accessible roof terrace. Distance to boundaries not demonstrable without a site plan.	MEDIUM	QIN DG Rev 2 p.97 (Sheet 6/7)
M-06	Balconies and terraces open to sky whose sides face neighbouring plots must have a 1.8 m high SOLID screen parapet.	Terraces at first and second floor face both side boundaries. No 1.8 m solid screen parapet is evident on any elevation.	MEDIUM	QIN DG Rev 2 p.92 (Openings)
I-01	Min 5% of plot as soft irrigated landscape; at least 2 mature trees in the front garden.	About 9 trees drawn along the front - likely satisfies the tree count. No landscape area schedule.	INFO REQUIRED	Plot sheet; QIN DG Rev 2 p.92
I-02	Swimming pool minimum 2.00 m from external plot boundary.	Pool drawn along the east side; not dimensioned to the boundary.	INFO REQUIRED	QIN DG Rev 2 p.92
I-03	Glazing U-value and SHGC step at 40% Window Wall Ratio. Roof U max 0.437, wall U max 0.568 W/m2K.	Extensive glazing to all elevations. No WWR calculation, no glazing specification.	INFO REQUIRED	KAHRAMAA Energy & Water Conservation Code, Issue 1, 29.05.2023
I-04	QCDD fire approval required at DC-1. Fire strategy, egress, ancillary separation.	No fire strategy drawing in the set.	INFO REQUIRED	QCDD Technical Requirements Guide 2022; QIN DG Rev 2 p.33
I-05	GSAS 2-star minimum, demonstrated at all CAC review stages.	Not evidenced in the set.	INFO REQUIRED	QIN DG Rev 2 p.32
I-06	Title block: PIN, plot and street numbers, applicable codes with editions, revision history.	No PIN, no revision number. Sheet dates inconsistent (A-100 13 June; others 18 May).	INFO REQUIRED	QCDD FF-01-2014-Rev03
P-01	Plot coverage max 50%.	Land 786.15 - free 458 = 328.15 m2 footprint = 41.7%.	COMPLIANT	Plot sheet; QIN DG Rev 2 p.92
P-02	Ground floor max +1.00 m above plot access level.	Ground floor +0.875 m.	COMPLIANT	QIN DG Rev 2 p.95
P-03	One basement level permitted.	One level at -2.10.	COMPLIANT	QIN DG Rev 2 p.92
P-04	Permitted land use - Residential Low Density Villa.	Single villa.	COMPLIANT	Plot Building Regulation Sheet
P-05	Plot identity and area must reconcile with the national cadastre.	Ministry of Municipality cadastre returns PIN 69052179, PDAREA 786 m2, PD/2807/2021. Plot coordinates compute to 786.16 m2. Drawings state 786.15 m2. All reconcile.	COMPLIANT	CGIS national cadastre, queried 10 Aug 2026
P-06	Minimum 2 on-plot parking spaces.	4 numbered bays shown at basement level.	COMPLIANT	Plot Building Regulation Sheet

3. Site geometry verification

Asked whether the building sits correctly within the plot limits, this review did not stop at 'no site plan provided'. The approved Plot Building Regulation Sheet carries the six plot corner coordinates on the Qatar National Grid. Those were used to reconstruct the plot and test the fit directly, and the plot identity was cross-checked against the Ministry of Municipality national cadastre.

Step 1 - Plot identity confirmed independently

Source	Plot area	Reference
National cadastre (CGIS / Ministry of Municipality), queried 10 Aug 2026	786 m2	PIN 69052179, PD/2807/2021
Plot corner coordinates, computed by NORMIA (shoelace)	786.16 m2	6 points, EPSG 2932
Plot Building Regulation Sheet, plan annotation	786.15 m2	Rev 1, 04 Aug 2022
Architect's title block, all 12 sheets	786.15 m2	LAND AREA

All four reconcile. The plot the drawings are based on is the correct one, and its area is verified against the national cadastre - not merely taken from the architect's title block.

Step 2 - Plot reconstructed from the coordinates

Property	Value
Shape	Trapezoid - two parallel long sides
Long sides (edges 3-4 and 6-1)	37.601 m and 37.603 m
Perpendicular distance between long sides	21.854 m
Extent along the long axis	38.142 m
Short ends	21.87 m (north-east, road frontage) and 19.95 m (south-west)
Perimeter	117.022 m

Step 3 - Buildable envelope after setbacks

Setback assignment taken from the annotations on the Plot Building Regulation Sheet: 5 m front to the north-east road frontage, 6 m rear to the south-west end, 3 m to each long side.

Dimension	Calculation	Available
Length	38.142 - 5.00 (front) - 6.00 (rear)	27.14 m
Width	21.854 - 3.00 - 3.00 (sides)	15.85 m
Envelope area	27.14 x 15.85	approx 430 m2

Step 4 - Fit test against the building as drawn

Building	Length	Width	Length margin	Width margin	Result
Structural grid 1-5 / A-D only	26.00 m	14.35 m	+1.14 m	+1.50 m	FITS
First floor as dimensioned on A-102	30.00 m	14.50 m	-2.86 m	+1.35 m	DOES NOT FIT

What this means

The structural core of the villa fits, with a little over a metre to spare in each direction. The **2.00 m first-floor projections beyond grids 1 and 5 are what break it** - they take the overall length to 30.00 m against 27.14 m available.

This conclusion does not depend on where the building is placed. A 30.00 m element cannot fit within a 27.14 m envelope at any position or offset. Shifting the building redistributes the encroachment between the front and rear setbacks; it cannot remove it.

Two qualifications, stated so the finding is not over-read: the test assumes the building's long axis runs parallel to the plot's long sides, which the plans indicate but do not dimension; and the width margin is calculated on the perpendicular distance between the parallel long sides, so it holds along the whole length. A dimensioned site plan remains required to confirm the actual position and to measure the front setback against its 10 m maximum.

Zoning context: the national GIS returns a single blanket 'SD - Special Development Area' polygon of about 1,185,333 m² over Qetaifan Island North, with no plot-level control. Development control here belongs to the master developer, which is why this review is run against the Plot Building Regulation Sheet and the Design Guidelines rather than the national R-zone regulations.

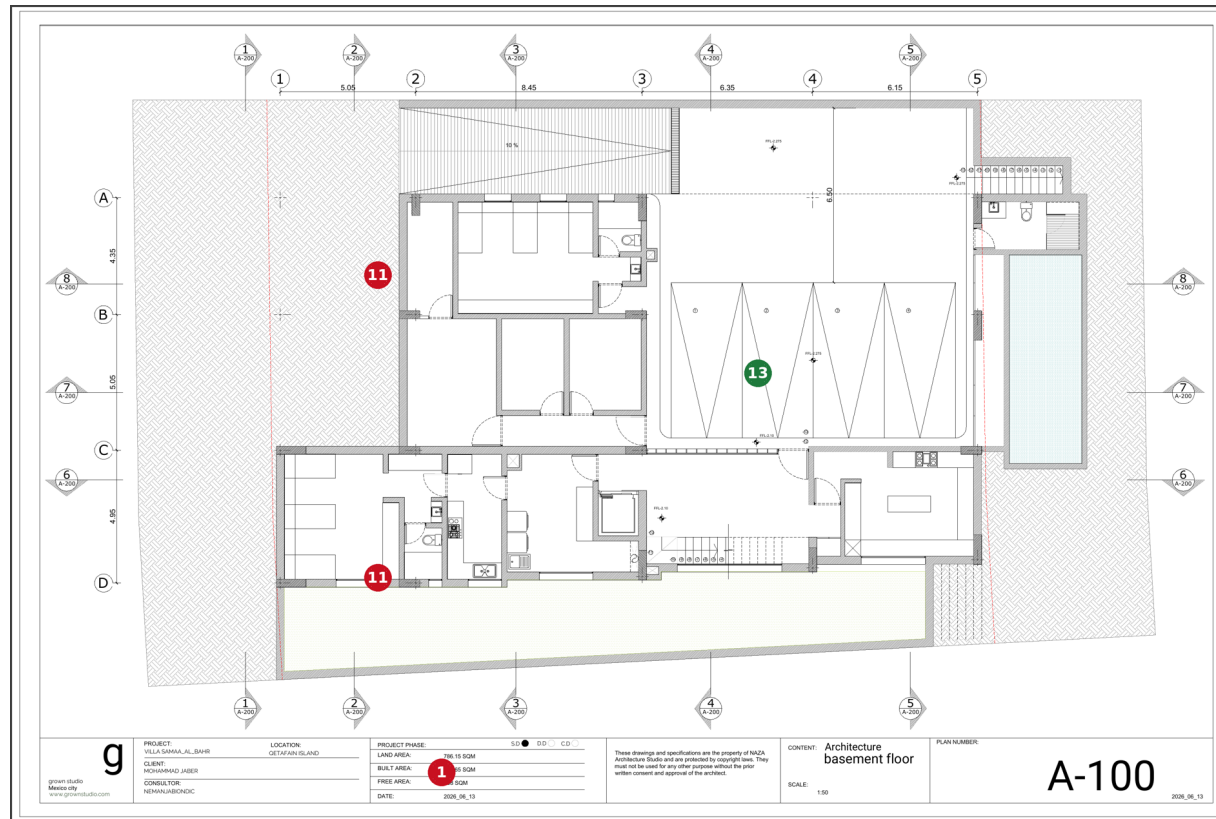
4. Sheet-by-sheet markup

Each drawing is reproduced with numbered callouts. Red = critical, amber = high, blue = information required. Callout numbers correspond to the key below.

No.	Severity	Finding
1	CRITICAL	Built area 819.65 m ² stated against a 472 m ² GFA cap.
2	CRITICAL	Top floor drawn and titled as a full second storey, not a 40% penthouse.
3	HIGH	First floor +6.825 and penthouse +10.675 exceed the +5.5 m / +9.0 m level bands.
4	HIGH	+15.00 measured from the access sidewalk, not from the lowest side of the plot.
5	HIGH	No site plan - setbacks to the plot boundary cannot be verified.
6	MEDIUM	2.00 m projections beyond the structural grid at both ends of the first floor.
7	MEDIUM	Boundary wall height not annotated; 2.5 m / 3.0 m conflict unresolved.
8	MEDIUM	Basement GFA exclusion unproven - no ventilation strategy shown.
9	INFO	Pool not dimensioned to the plot boundary (2.00 m minimum).

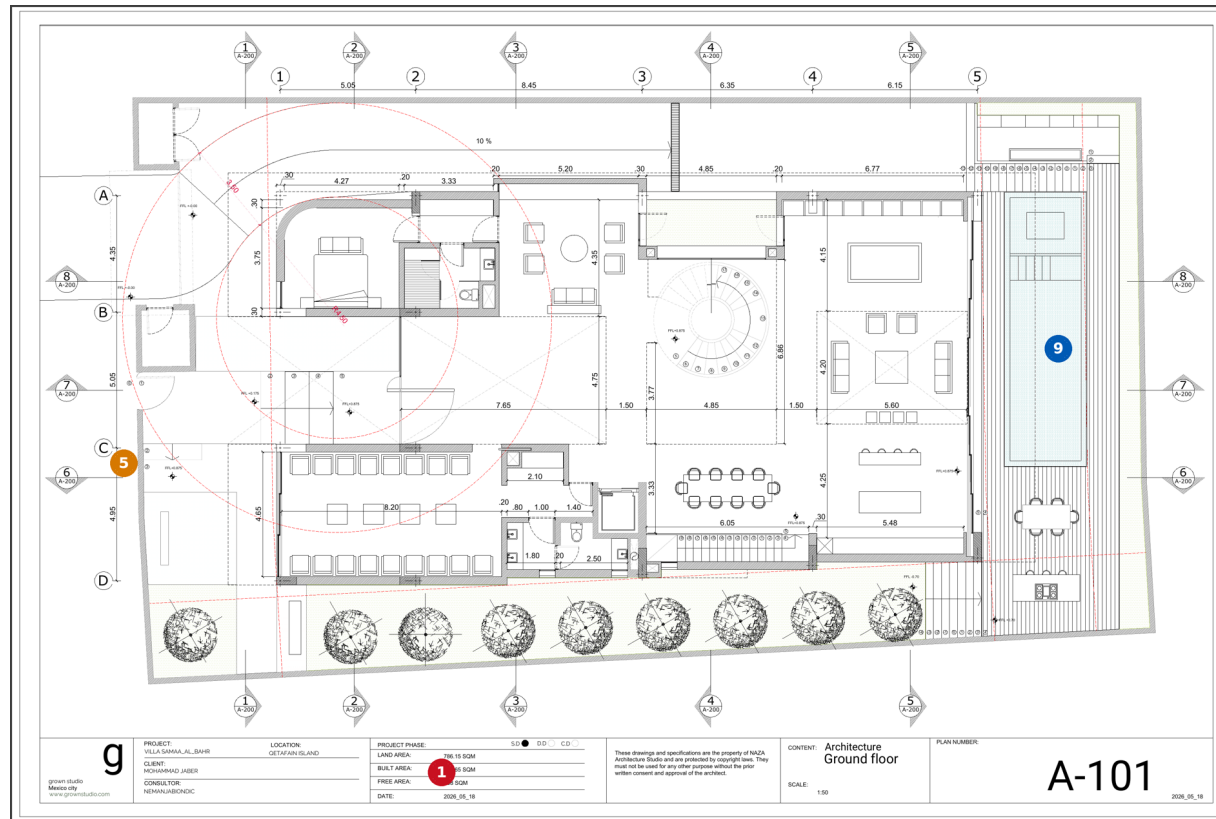
No.	Severity	Finding
10	INFO	No WWR calculation or glazing specification against the KAHRAMAA bands.
11	CRITICAL	Habitable rooms at basement level - bedrooms, bathrooms, kitchen, living.
12	CRITICAL	Occupied roof terrace at FFL +15.00 - at the height limit, before any parapet.
13	COMPLIANT	4 parking bays shown, exceeding the 2-space minimum.

A-100 - Basement floor



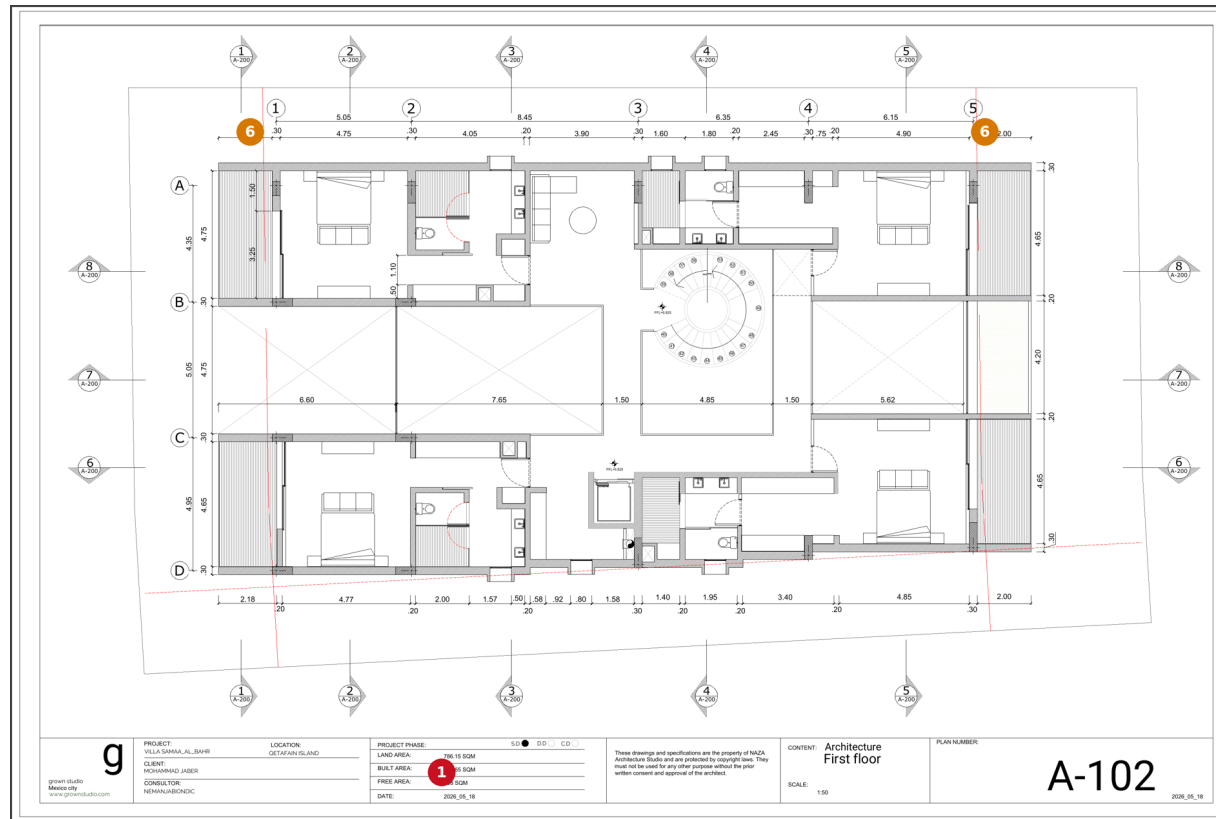
- 1 - Built area 819.65 m2 stated in the title block against a 472 m2 cap.
- 11 - CRITICAL. The basement contains BEDROOMS, BATHROOMS, A KITCHEN and LIVING SPACE. GFA exclusion requires the area to be BOTH naturally ventilated AND non-habitable. A habitable basement counts toward the 472 m2.
- 11 - The Design Guidelines also cap habitable basement area at 50% of the basement footprint. On inspection the habitable portion looks close to or above that; it needs measuring.
- 13 - CORRECTION TO THIS REVIEW: 4 numbered parking bays and a 10% ramp are shown. The 2-space minimum is exceeded. Bay dimensions and aisle width are still unannotated.
- Single basement level at -2.10 / FFL -2.275 is compliant in number.
- The basement outline appears to extend beyond the above-ground footprint toward the east boundary - the basement must follow the BUILDING setback.

A-101 - Ground floor



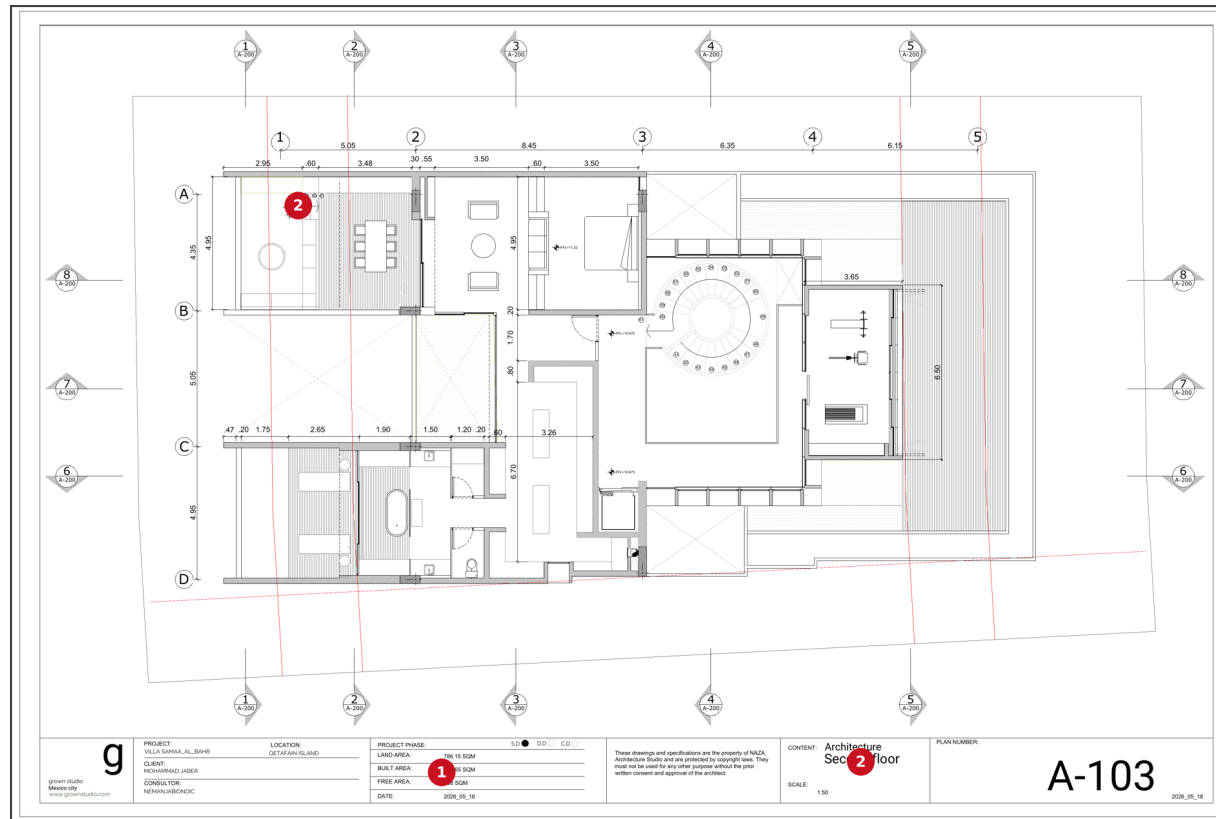
- 1 - Built area discrepancy repeated in the title block.
- 5 - No site plan exists in the set; this is the only sheet showing the plot outline, and it carries no setback dimensions to the boundary.
- 9 - Pool drawn along the east side, not dimensioned to the plot boundary (2.00 m minimum applies).
- Ground floor FFL +0.875 is COMPLIANT against the +1.00 m maximum above access level.
- Approximately 9 trees along the front frontage - the 2 mature tree requirement is likely met, but no landscape area schedule is provided.

A-102 - First floor



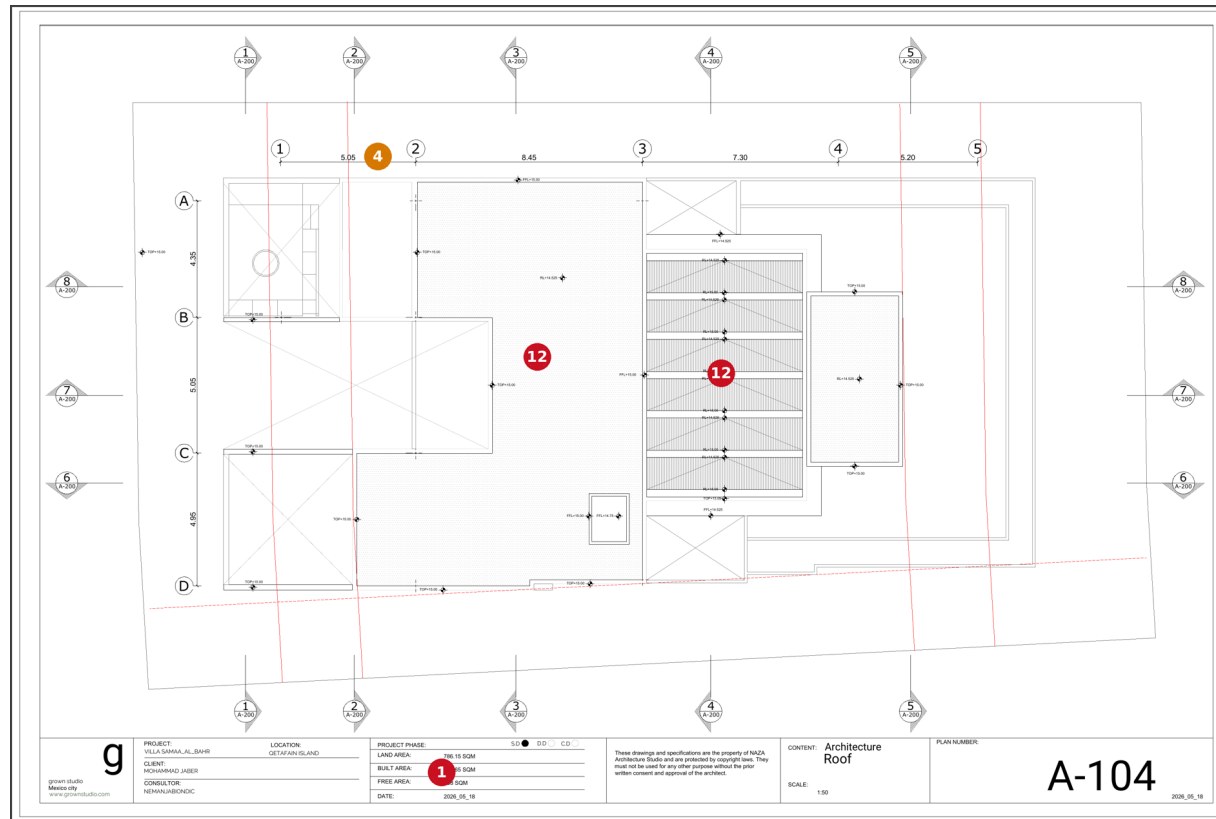
- 1 - Built area discrepancy repeated.
- 6 - Overall dimension strings total 30.00 m against a 26.00 m structural grid, indicating 2.00 m projections beyond grids 1 and 5 at both ends.
- Only architectural elements may project 1.5 m into the neighbour side setback, and they may carry no openings. These read as habitable/terrace projections.
- First floor FFL +6.825 exceeds the +5.5 m level band (finding 3).

A-103 - Second floor



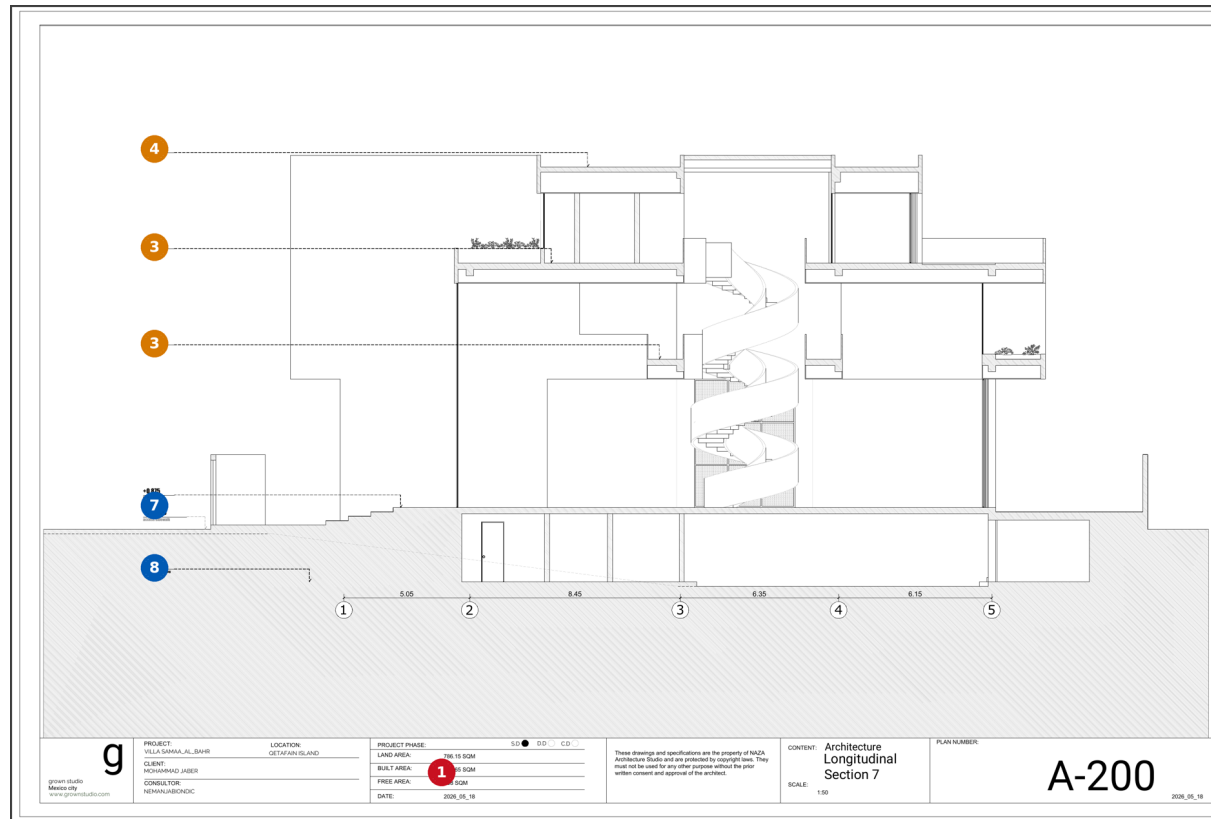
- 2 - CRITICAL. This sheet is titled 'Second floor'. The plot permits G+1+P only.
- 2 - The level contains bedrooms, bathrooms, living space, a kitchen and terraces distributed across both ends of the plan - materially more than 40% of the first-floor footprint, with no consistent 3 m setback from the first-floor front and rear facades.
- The sections label this same level 'penthouse'. The set is internally inconsistent about what this storey is.
- FFL +10.675 (also +10.15, +11.22) exceeds the +9.0 m level band.

A-104 - Roof



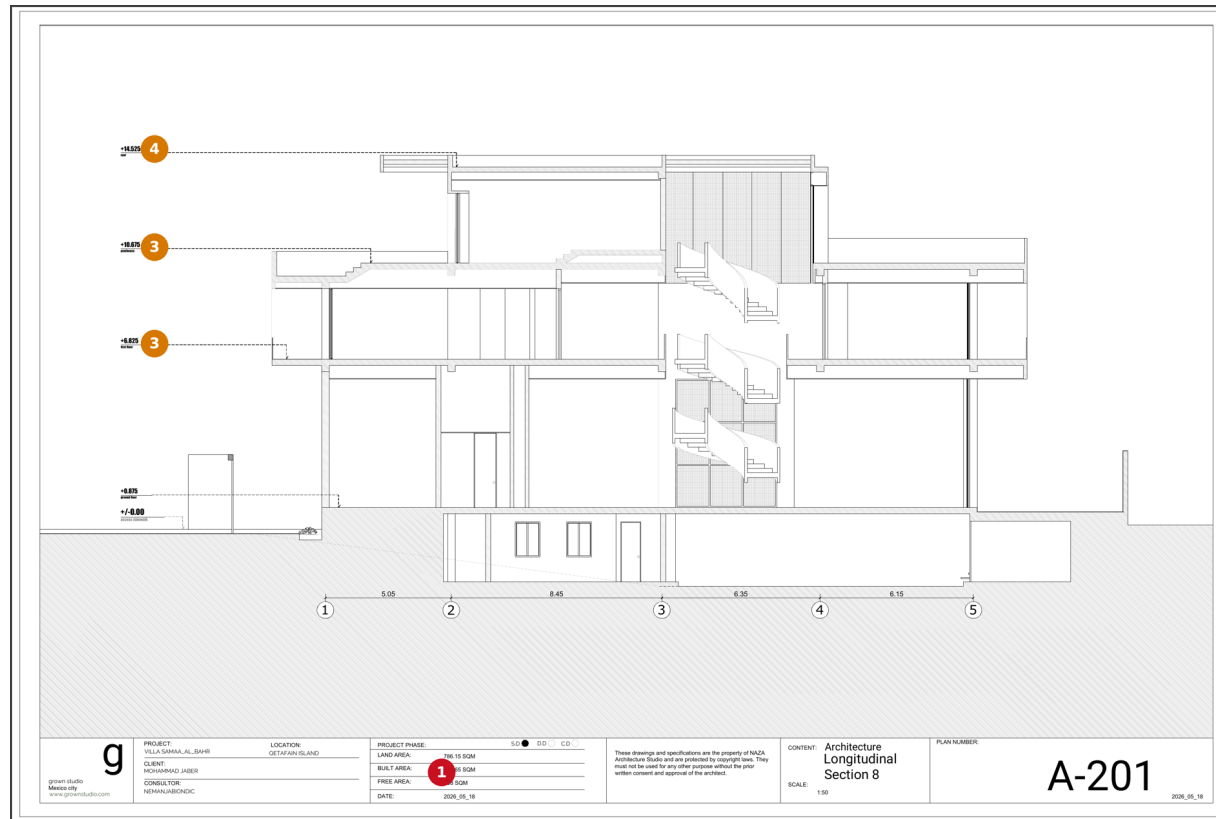
- 12 - CRITICAL. This is not a bare roof. FFL +15.00 and FFL +14.75 denote an OCCUPIED roof terrace, with a pergola / louvred structure at RL +14.525 to +15.00.
- 12 - A finished floor level AT +15.00 cannot carry a parapet, balustrade or guarding without exceeding the 15 m limit - which expressly includes all parapets and roof structures. The Design Guidelines show screens concealing roof structures sitting BELOW the +15 m line, not above it.
- 4 - TOP +15.00 and RL +15.00 recur across the whole roof; the building sits exactly at the limit with zero headroom.
- The limit is measured from the LOWEST SIDE OF THE PLOT, not the access sidewalk used here. Any fall across the plot puts the building over.
- Roof terraces are permitted only where at least 6 m from side and rear plot boundaries, or adequately screened. Neither is demonstrable without a site plan.

A-200 - Longitudinal Section 7



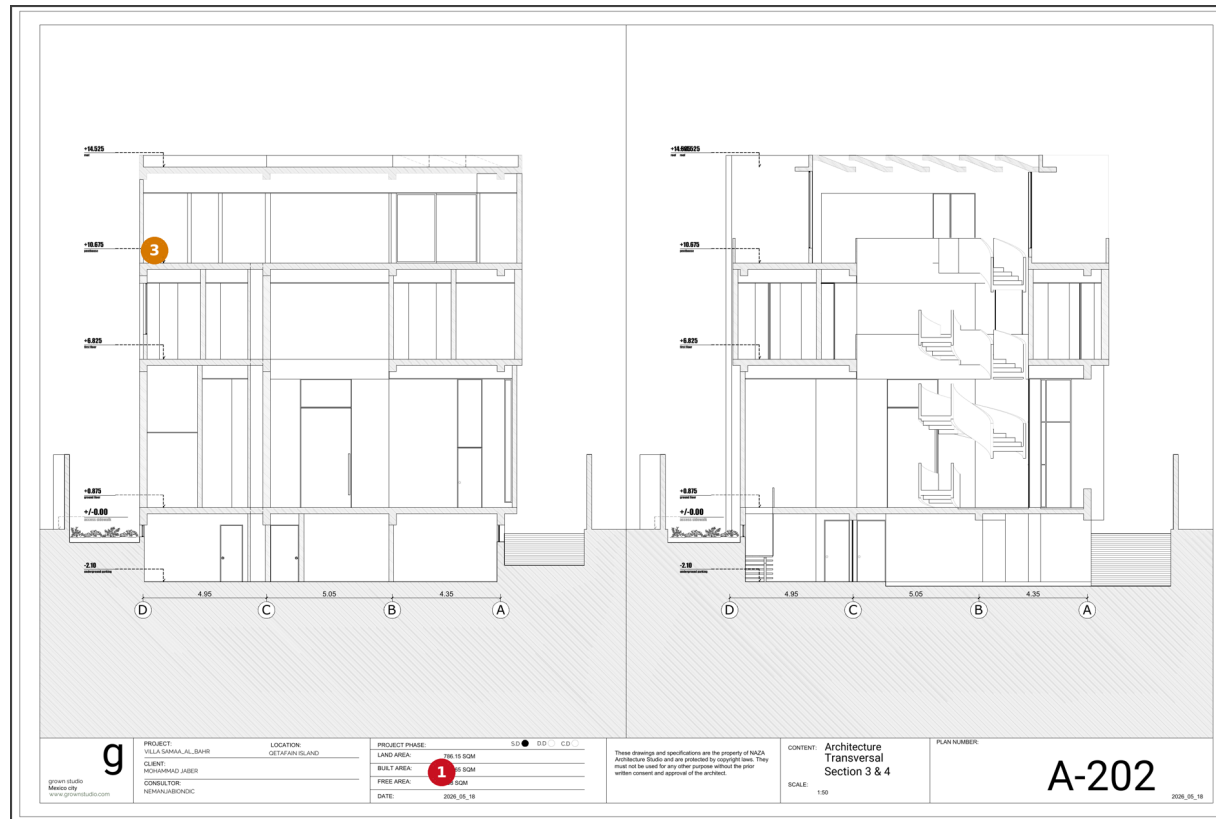
- 3 - The level stack is the clearest evidence of the height findings: ground +0.875, first +6.825, penthouse +10.675, roof +14.525.
- Ground-to-first storey height is 5.95 m; first-to-penthouse 3.85 m. Generous storey heights are what push the intermediate levels past the +5.5 m and +9.0 m bands.
- 4 - Roof at +14.525 with elements to +15.00.
- 7 - Boundary wall visible at left and right but not annotated with a height.
- 8 - Underground parking at -2.10; no ventilation strategy shown.

A-201 - Longitudinal Section 8



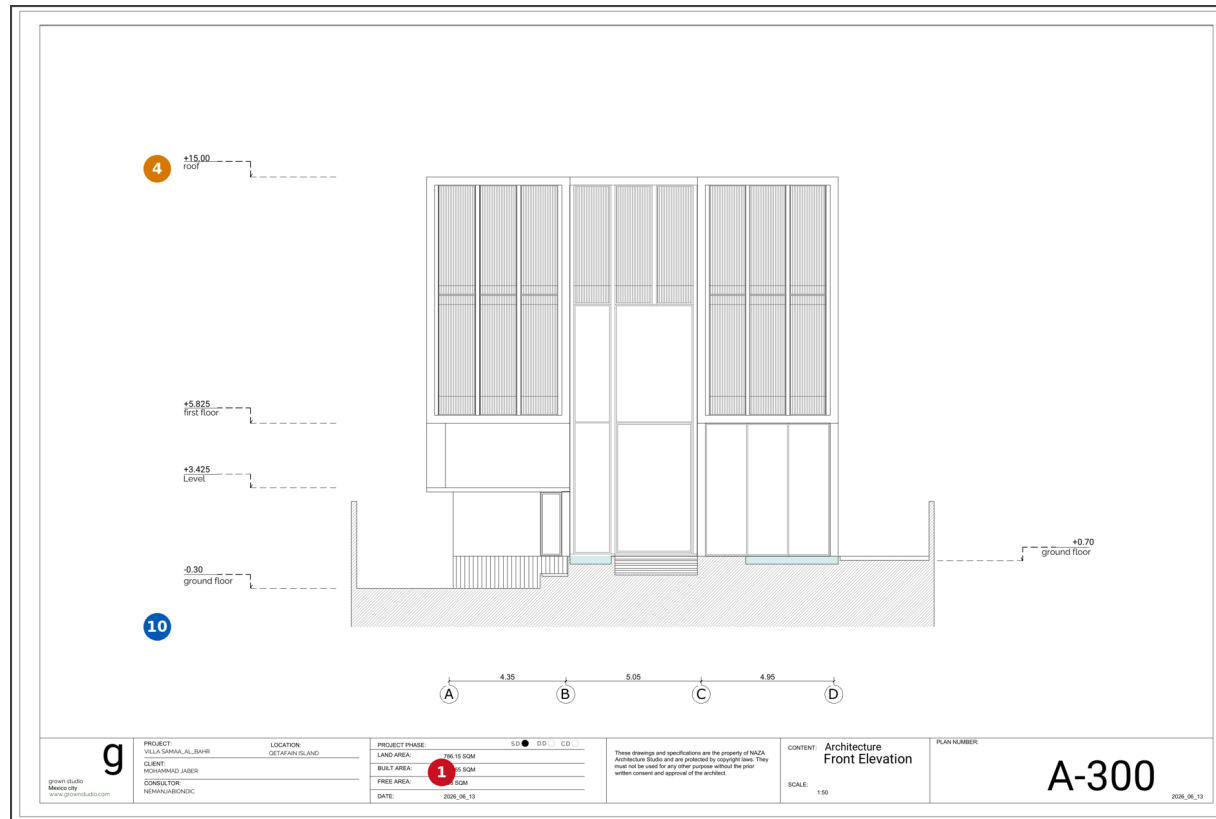
- 3 / 4 - Same level stack and datum issue as A-200.
- No basement shown on this section line.

A-202 - Transversal Sections 3 & 4



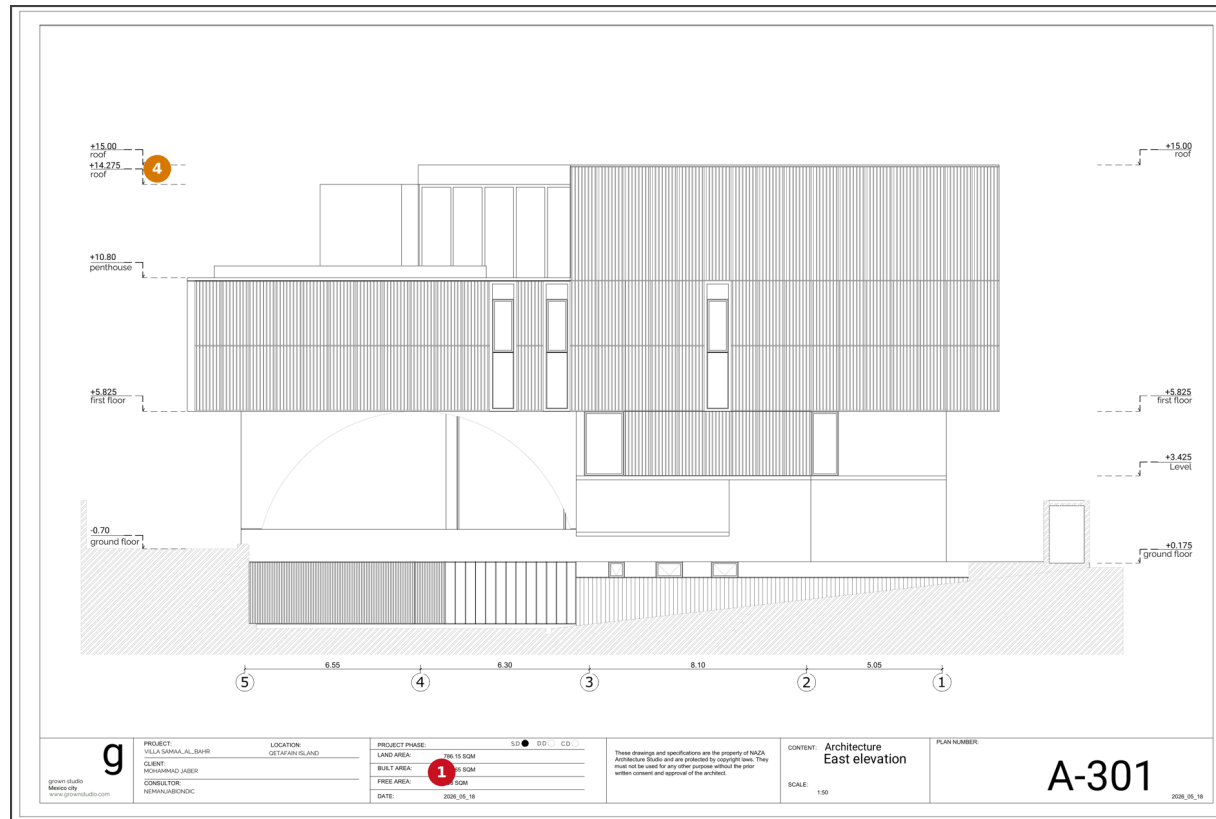
- 3 - Confirms the level stack across the short axis.
- Transverse sections would be the natural place to show setback dimensions to the side boundaries - none are given.

A-300 - Front Elevation



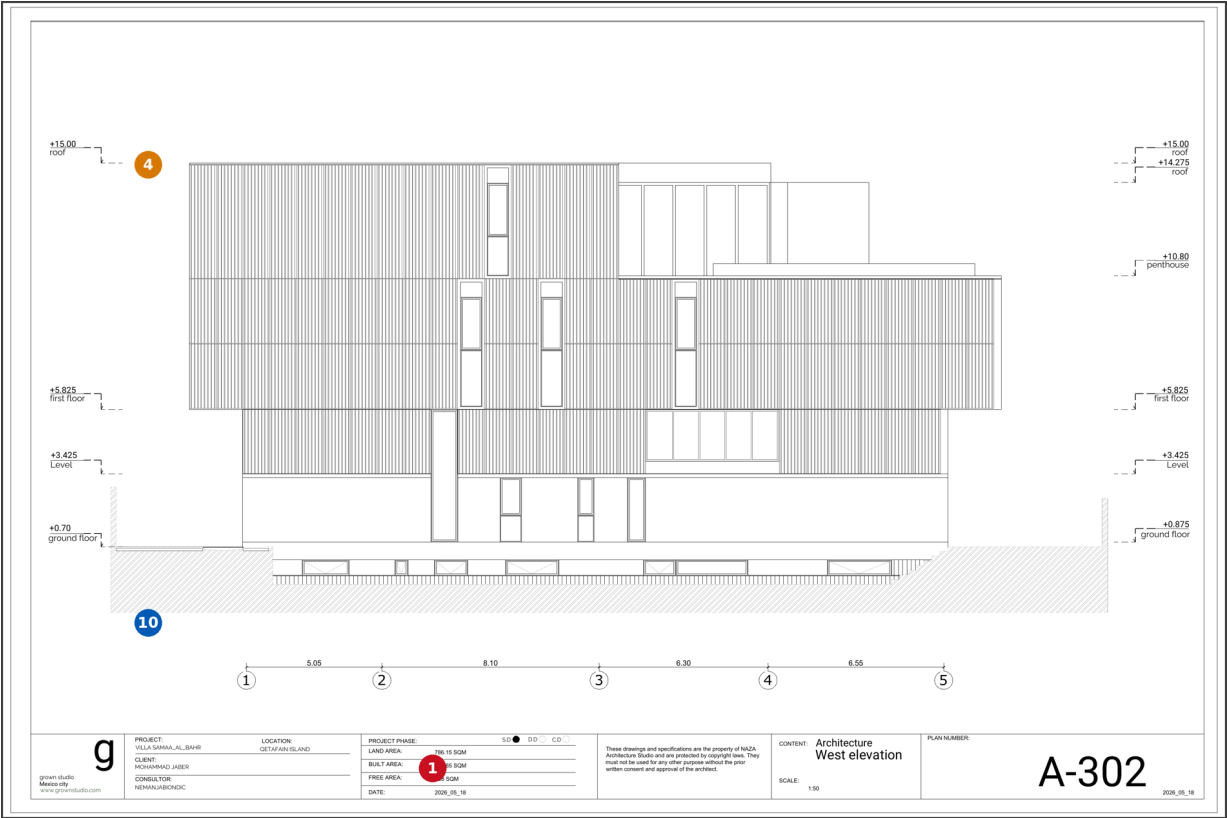
- 4 - Elevation tops at +15.00, exactly at the limit.
- 10 - Extensive glazing with no Window Wall Ratio calculation. The KAHRAMAA requirement steps at 40% WWR: below it, U 3.30 / SHGC 0.30; above it, U 2.10 / SHGC 0.25.
- Boundary wall height not annotated on the street elevation - the most visible element of the villa from the public realm.

A-301 - East elevation



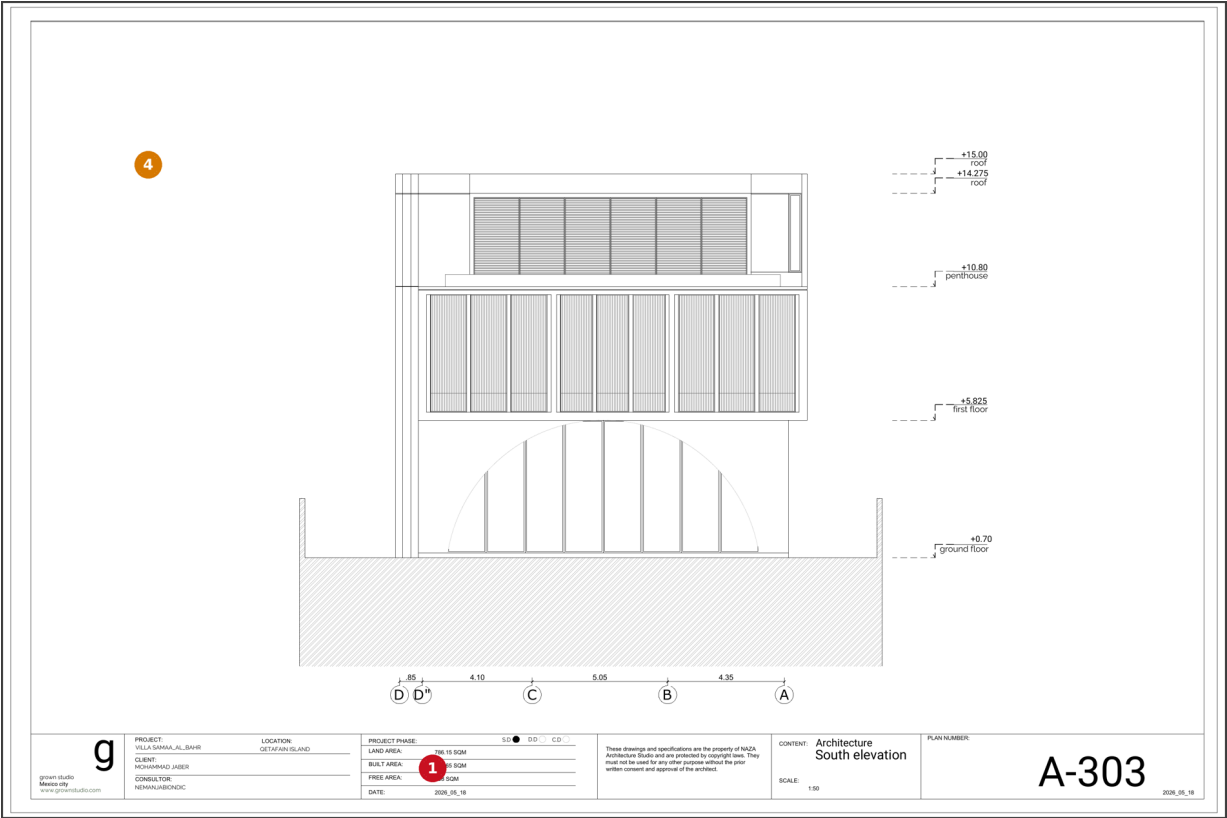
- 4 - +15.00 / +14.275 / +10.80 / +5.825 confirm the overall height and the upper level positions.
- No boundary wall or plot boundary shown.

A-302 - West elevation



- 4 - Same height condition.
- 10 - Glazing extent again unquantified.

A-303 - South elevation



- 4 - Same height condition.
- Second floor reads as a full storey in elevation, supporting finding 2.

5. How to change the design

This section sets out what has to change, in the order it has to change. Items are sequenced so that resolving the earlier ones does not force rework of the later ones.

Step 1 - Establish the true GFA position (before anything else)

Produce an area schedule by floor stating gross area, what is counted, and what basement area is claimed as excluded. Until that exists, nobody knows how far over the scheme is. The arithmetic that matters: footprint is 328.15 m² (land 786.15 minus free area 458). Three above-ground levels at that footprint would be roughly 984 m² against a 472 m² cap. The stated 819.65 m² sits between those, so the upper levels are already reduced - but not nearly enough.

Target: 472 m² above ground, inclusive of every ancillary structure. On a 328 m² footprint that is approximately one and a half floors of full-footprint area. The scheme as drawn is roughly a third over even on the most favourable basement assumption.

Step 2 - Decide what the top level is, and commit to it

Option	What it means	Consequence
A. True penthouse	Reduce the top level to a maximum of 40% of the first-floor footprint, inset at least 3 m from the first-floor front and rear facades. Retitle the sheet 'Penthouse'.	Recovers a large part of the GFA overrun in one move. Any penthouse elevation facing a neighbour must be 6 m from the boundary wall to carry openings - so expect blind flanks unless the volume is pulled in further.
B. Two storeys only	Delete the third level entirely and redistribute accommodation across ground and first.	Simplest route to compliance on storey count. Costs the roof-level living space and the sea views that presumably motivated it.

Option A is the one worth pursuing, because it addresses C-01 and C-02 together.

Step 3 - Reset the section

- Re-datum the whole section to the LOWEST SIDE OF THE PLOT, not the access sidewalk. Obtain the plot road levels first - the levels drawing currently in the project set is a site-wide 1:500 sheet covering many plots and does not resolve this one.
- Bring the first floor down within the +5.5 m band. The current 5.95 m ground-floor storey height is the direct cause; approximately 4.6 m floor-to-floor would sit inside the band.
- Bring the penthouse level down within the +9.0 m band - currently 1.675 m over.
- Re-check the 15 m overall after re-datuming. There is currently zero headroom, so any fall across the plot creates a breach.
- Allow for roof plant screening WITHIN the 15 m, not above it - the Design Guidelines show screens concealing roof structures below the +15 m line.

Step 4 - Produce the missing drawings

- **Site plan** - dimensioned to the plot boundary on all six sides, showing setbacks (5 m front / 3 m sides / 6 m rear, and the maximum 10 m front), the boundary wall, gates, pool, ancillary structures, driveway and the vehicular and pedestrian access points fixed by the plot sheet.
- **Area schedule** by floor with the GFA basis stated.
- **Parking layout** - minimum 2 spaces, bays 2.65 x 5.80 m, aisle 6.70 m.
- **Boundary wall elevations** - design to 2.50 m at the front until the conflict is ruled on, so that either outcome remains buildable.

- **Landscape plan** evidencing 5% soft irrigated area and the 2 mature front-garden trees.
- **Fire strategy** - required for QCDD approval at DC-1, in parallel with the architectural review, not after it.

Step 5 - Envelope and services

- Calculate Window Wall Ratio per elevation and decide deliberately whether to sit below 40% or above it. The KAHRAMAA requirement steps at that threshold - below: U 3.30, SHGC 0.30; above: U 2.10, SHGC 0.25. Sliding from 38% to 43% during design development without re-specifying the glass is a route to a refused electricity connection.
- Specify roof and wall build-ups to U 0.437 and U 0.568 W/m²K. The roof requirement is the tighter of the two and will drive insulation depth - which interacts with the 15 m limit.
- Confirm the basement natural ventilation strategy in a form that satisfies the GFA exclusion test, or accept the area into the GFA count.
- All surface, storm and roof water must be disposed of within the plot - not to the road, the neighbours or the foul sewer. Coordinate with the basement footprint and the soft landscape area.
- Appoint a GSAS service provider now. Orientation, massing, shading and envelope decide several criteria and are expensive to recover later.

Step 6 - Put the open questions to the authority in writing

- Is the November 2025 Front Boundary Wall document binding on this plot, and does it supersede the 3.00 m height on the approved Plot Building Regulation Sheet?
- What is the quantitative test for 'naturally ventilated' for basement GFA exclusion?
- What governs fire and life safety for villas, given the 2022 QCDD guide supersedes the 2015 guidelines but contains no dwelling chapter?
- Which accessibility standard is accepted for villa submissions?
- Confirm the surveyed lowest point of the plot for the height datum.

6. What this review could not test

Stated so the result is not over-read.

Not tested	Why
Exact building position on the plot	Tested by reconstruction from the plot coordinates (Section 3), which proves the building cannot fit. A dimensioned site plan is still needed to confirm placement and to measure the front setback against its 10 m maximum.
Exact penthouse ratio	No area schedule. The finding rests on the sheet title and the drawn extent, which is sufficient to raise it but not to quantify it.
Height against the true datum	Plot-specific road levels not available; the supplied levels drawing is a site-wide 1:500 sheet.
Construction technology - waterproofing, insulation build-up, finishes	No Qatar Construction Specifications clause content held. The official QCS portal could not be reached, and third-party copies have no edition control.
Fire compliance for a villa	The 2022 QCDD guide has no dwelling or villa chapter, and the 2015 provisions it supersedes have not been replaced.
Accessibility	Qatar has no confirmed national built-environment accessibility code. Law No. 22 of 2025 creates a duty; its executive regulations were not confirmed as issued.
Structural and MEP	Not included in the submitted set.

Golden rule

Where sufficient evidence exists, NORMIA reports VERIFIED COMPLIANT. Where a violation is demonstrated, NON-COMPLIANT. Where information is incomplete, INSUFFICIENT INFORMATION. Where a requirement cannot be verified, UNVERIFIED - AUTHORITY CONFIRMATION REQUIRED. No regulation in this report has been inferred or invented; every finding cites the document, edition and page it was tested against.